

HEALTHCONNECT CLINIC EXPERIENCE LAB

Project Title: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Track: Data Analytics

Week 7: Testing, Refinement and End-to-End Validation

Week 6 to Week 7 Transition

Week 7 continued directly from the Week 6 advanced analytics and Power BI dashboard. The main Week 6 output was an improved analytical dashboard containing validated KPIs, segmented no-show analysis, a risk heatmap and an evidence table.

The main Week 6 findings showed that previous no-shows and long booking lead times were strongly associated with missed appointments. Appointments booked 31+ days in advance had a **60.49% no-show rate**, while patients with 2+ previous no-shows had a **61.02% no-show rate**. The combined group of patients with 2+ previous no-shows and 31+ booking lead days recorded a **72.96% no-show rate**.

Week 7 therefore focused on testing the accuracy of these outputs, validating dashboard calculations, checking filters, resolving inconsistencies and refining the interpretation of the high-risk segment.

Testing Approach

The Power BI dashboard and its underlying calculations were tested against the HealthConnect appointment dataset. The main objectives were to confirm that KPI values were accurate, dashboard results agreed with the underlying data, filters worked correctly, and important Week 6 findings remained valid.

The testing included KPI validation, segmented analysis, dashboard filter testing, validation of important findings and refinement of visualisations. Results were compared with the underlying 5,000 appointment records.

KPI Validation

The main KPIs were successfully validated. The dataset contains **5,000 appointments**, consisting of **2,314 attended appointments**, **2,423 no-shows** and **263 cancellations**. This produces an attendance rate of **46.28%**, a no-show rate of **48.46%** and a cancellation rate of **5.26%**.

All major KPI calculations matched the expected results and therefore passed testing.

The important segmented results were also validated. The 31+ booking lead group recorded a **60.49% no-show rate**, while the group with 2+ previous no-shows recorded **61.02%**.

The combined high-risk segment was clarified during testing. It was defined as:

Previous No-Shows ≥ 2 AND Booking Lead Days ≥ 31

This group contained **233 appointments**, of which **170 were no-shows**, producing a validated no-show rate of **72.96%**.

Dashboard Testing and Refinement

The major dashboard filters were tested, including date, appointment type, gender, age group, appointment day and reminder status. The dashboard responded consistently to the selected filters and recalculated the displayed results appropriately.

The dashboard was refined to make the high-risk segment clearer and reduce confusion between individual risk factors and the combined high-risk group. The evidence table was retained because it shows the appointment count, no-show count and no-show rate behind important findings.

The Week 6 heatmap comparing previous no-shows with booking lead time was also retained because it clearly demonstrates how the two factors interact.

Testing, Refinement and Validation Results

Component Tested	Expected Result	Actual Result	Status	Action
Total Appointments	5,000	5,000	PASS	None
No-Show Rate	48.46%	48.46%	PASS	None
Attendance Rate	46.28%	46.28%	PASS	None
Cancellation Rate	5.26%	5.26%	PASS	None
Lead 31+ No-Show Rate	60.49%	60.49%	PASS	None
Previous No-Shows 2+ Rate	61.02%	61.02%	PASS	None
Combined High-Risk Segment	72.96%	72.96%	PASS	Definition clarified

The testing confirmed that the main analytical calculations were accurate. The only important refinement was clarifying the definition of the combined high-risk segment.

Validation of Important Findings

The Week 6 findings remained consistent after testing. Previous no-shows continued to show a strong association with future missed appointments. The no-show rate increased from **43.51%** among patients with no previous no-shows to **68.82%** among those with 3+ previous no-shows.

Booking lead time also remained an important indicator. The no-show rate increased from **26.51%** for appointments booked 0–1 days ahead to **60.49%** for appointments booked 31+ days ahead.

Reminder status showed a lower observed no-show rate when a reminder was sent (**47.36%**) compared with when no reminder was recorded (**51.39%**). However, this is an association and does not prove that reminders directly caused the difference.

Distance was also retained as a relevant factor because appointments more than 20 km from the clinic recorded a **57.76% no-show rate**.

Cross-Track Testing

The Data Analytics track collaborated with **Data Science** to validate analytical variables that could support future predictive modelling.

The variables reviewed included previous no-shows, booking lead time, reminder status, reminder channel, appointment type and distance to clinic. The validated findings were shared as potential predictive features.

The testing also highlighted the importance of avoiding data leakage by ensuring that variables used for prediction are available at the appropriate point in the appointment process.

This collaboration resulted in a meaningful testing process involving validation of the analytical findings, identification of potential modelling considerations, refinement of variable selection and confirmation of the outputs.

Remaining Limitations and Risks

Some limitations remain. The dataset is fictional and anonymised, so the findings should not automatically be treated as representative of real-world patient behaviour. The analysis identifies associations rather than causation.

There are **90 missing distance values and 60 missing waiting-time values**, which limit analysis of these variables. Reminder effectiveness also requires further investigation before causal conclusions can be made.

The dataset contains Sunday appointments even though the HealthConnect Knowledge Base indicates that the clinic is closed on Sundays. Therefore, Sunday patterns should not be used for major operational decisions until the inconsistency is clarified.

Updated Recommendations

HealthConnect should consider targeted, non-punitive support for patients with repeated previous no-shows. Appointments booked 31+ days in advance should receive appropriate reminder and follow-up support.

The combined high-risk group of patients with 2+ previous no-shows and 31+ booking lead days should receive particular attention because its validated no-show rate was **72.96%**.

Reminder coverage should continue to be monitored, while further testing should investigate reminder timing and channel effectiveness. Distance-related barriers should also be investigated.

Week 8 Readiness

Before final integration, the Power BI dashboard and validated KPI calculations should be finalised. The high-risk definition should remain consistent across the final report and presentation. The Sunday data inconsistency should be clarified, and remaining cross-track dependencies should be confirmed.

Overall, Week 7 confirmed that the Data Analytics component is functioning as intended after testing and refinement. The main KPIs and important analytical findings were successfully validated, the high-risk segment was clearly defined, and the dashboard was improved for clearer decision-making. The validated outputs are ready to contribute to the final HealthConnect integration and presentation in Week 8.